NLP Sentiment Analysis

CptS 440

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Introduction

Sentiment analysis is a Natural Language Processing task that focuses on determining the emotional tone or opinion expressed in a piece of text. In many real world settings, large amounts of text data are created through product reviews, social media posts, customer feedback forms, and online discussions. Manually reading and labeling this text is time consuming, so automated sentiment analysis systems are useful for quickly identifying whether a piece of text expresses a positive or negative opinion.

For this project, our goal was to build a sentiment analysis system that classifies text as either positive or negative. We focused on binary sentiment classification because the primary dataset used in our experiments was an IMDB Movie Reviews dataset that was labeled using positive and negative review categories. Instead of building only one model, we implemented and compared multiple machine learning approaches to understand how different model types perform on the same task. Our final system compares three main models: a Convolutional Neural Network (CNN), Naive Bayes, and Gradient Boosting. These models represent different approaches to text classification. Naive Bayes provides a simple probabilistic baseline, the Gradient Boosting model provides a traditional machine learning comparison, and the CNN provides a neural network based approach that can learn patterns from text. We evaluated the models using accuracy, precision, recall, F1 score, confusion matrices, and ROC curves.
The main purpose of the project was not only to produce sentiment predictions, but also to compare the strengths and weaknesses of each approach. This allowed us to examine whether a more complex model like CNN had more advantages over simpler models like Naive Bayes.

Dataset & Preprocessing

The primary dataset used for training and evaluating our models was the IMDBReviews.csv dataset, consisting of 50,000 labeled reviews evenly split between positive and negative sentiment. This dataset has clear binary ‘positive’ or ‘negative’ labeling for corresponding positive and negative reviews. This served as a good benchmark to train our models for the purpose of sentiment analysis. In addition, the Yelp Business dataset was used in the project's classification application, allowing trained models to perform live sentiment analysis on real Yelp customer reviews. This dataset provided our team with real world customer reviews with complex and sometimes ambiguous reviews for our trained models to classify and label. Before training, all text was passed through a shared preprocessing pipeline:

1. **Contraction expansion** — Contractions are expanded, “don't” becomes “do not” ect., using the contractions library, so negations are not obscured
2. **HTML tag removal** — Residual characters from web-scraped text is stripped
3. **Negation tagging** — Words within a four word phrase range with a negation,”not”, “never” etc., are concatenated with a _NEG suffix (e.g. "not" becomes "not_NEG". This

step preserves sentiment that would otherwise be lost during stopword removal, a phrase like "not good" could be reduced to “good” for example

4. **Non-alphabetic character removal** — punctuation and digits are stripped with the exception of _NEG tags
5. **spaCy NLP filtering** — Remaining text is processed and lemmatized through spaCy's en_core_web_sm pipeline, “running” becomes “run”, better becomes “good” ect., and part-of-speech filtering, retaining only nouns, verbs, adjectives, and adverbs. Stopwords are removed, with negation words excluded from removal to preserve _NEG tags

The processed dataset was then split into training and test sets using a 80/20 ratio, ensuring consistent class distribution. The training set was used exclusively for model fitting, while the test set was held out for final evaluation to measure generalization.

Naive Bayes

Naive Bayes is a probabilistic classification model based on Bayes’ theorem. In the context of sentiment analysis, the model estimates the probability that a review is positive or negative given the words that appear in the review. For example, if words such as “amazing”, “excellent”, or “masterpiece” frequently appear in positive reviews, the model learns that those words increase the probability of a positive label. Similarly, if words such as “terrible”, “boring”, or “waste” frequently appear in negative reviews, the model learns that they increase the probability of a negative label. In our project, Naive Bayes used TF-IDF features. TF-IDF converts text into numerical vectors based on how important a word is in a document relative to the full dataset. Words that appear often in one document but are not overly common across all documents receive higher weight. This helps the model focus on more informative words and locate and identify important keywords within the text.

Gradient Boosting

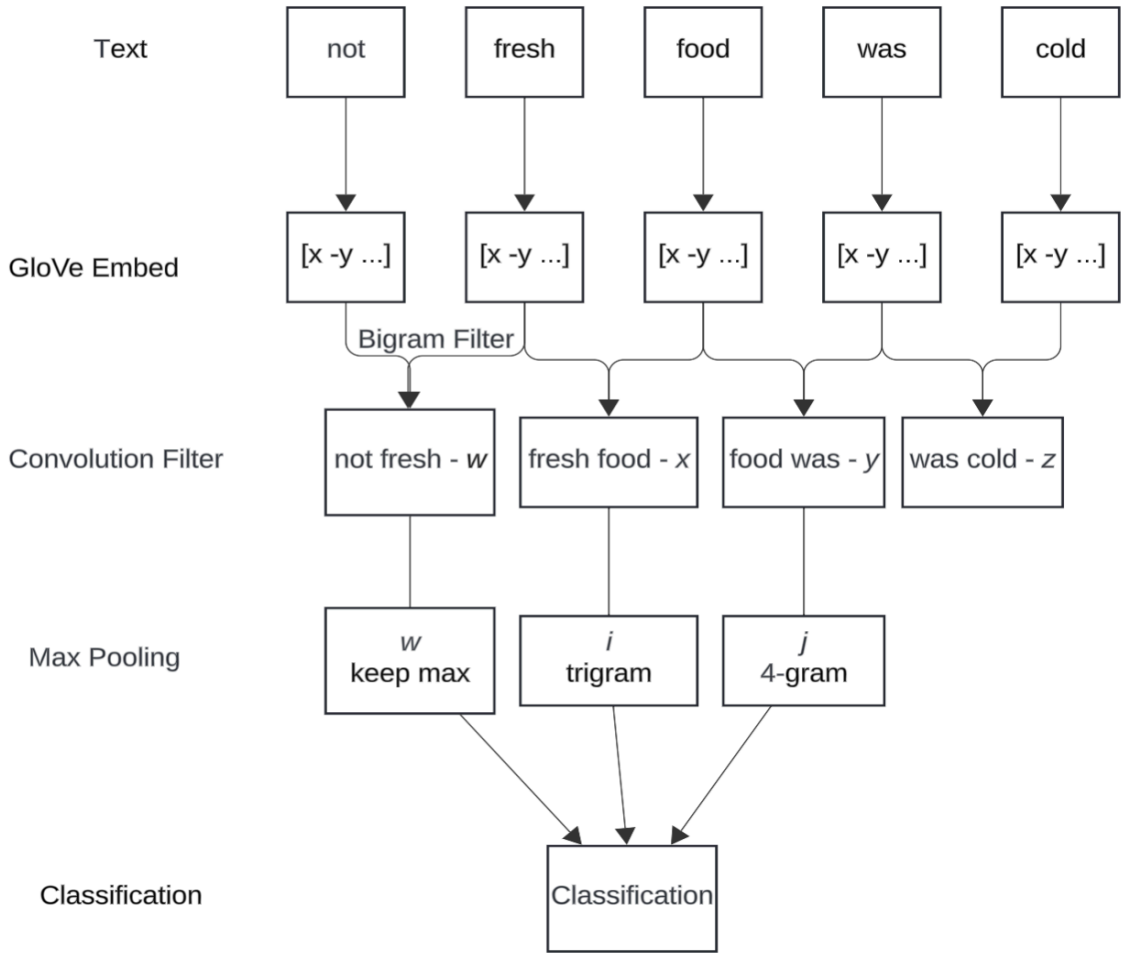
Our group used gradient boosting as a sanity check to our other models, as well as a way to determine ambiguity between words. By using Gradient Boosting alongside CNN’s during business analysis, the Gradient Booster can determine ambiguity within a given piece of text by having a disagreement between the Boosters sentiment analysis and the CNN’s. Because the Gradient Booster treats words independently, discretely classifying each in an ensemble manner, any difference in classification between the Booster and the CNN indicates a point of ambiguity in the review. Due to the drastically different methods the two models use, when there are two different classifications between the models there is usually some sort of ambiguity contained within the review. Using 200 trees of depth 4, the Gradient Booster achieves an accurate classification rate to help our team search for ambiguity inside the Yelp reviews.

Convolutional Neural Network

Convolitional Neural Networks are Neural Networks usually used for computer vision. That is translating photos or images into information that an algorithm or model can understand and

manipulate. Our group was able to utilize the CNN model in such a way that we are able to leverage the ability of the CNN model's ability to associate and identify images to measure user sentiment. Traditionally a CNN model would move filters across an image detecting edges and shapes. Our group utilized this filter to instead classify sentiment over text, instead of pixels the CNN is filtering embedded word vectors in text files. We constructed our model to filter for and recognize sentiment containing phrases. Combinations of words like "very enjoyable", and "not good" are discerned by the CNN where it would traditionally recognize an edge in an image for example. This is achieved through the use of GloVe (Global Vectors for Word Representation), a set of pre-trained word embeddings, with each word encoded with a semantic meaning. These word embeddings are passed through parallel convolutional filters of sizes 2,3, and 4. Using maxpooling the strongest positive or negative signal from each signal is selected and combined to generate an associated sentiment for the phrase or word. This generated sentiment is then compiled in the context of the analyzed text document, from where the text can then be classified as 'negative' or 'positive'.

The glove embedding GloVe our group used contained 100 dimensions per word [n_1, … , n_100]. These embeddings are then used to create a matrix composed of the phrases chosen by the CNN. The data flow for the CNN would look like:



With *w,x,y,z* and *i,j* representing some continuous values in each word embed determined by the GloVe word vectors, the chart above demonstrates how the CNN model is able to classify words and phrases. The CNN model also proved to be the best way to classify the Yelp reviews due to its ability to contextualize words and phrases in the text, and as such was our main tool in classifying the yelp reviews.

Evaluation Metrics

To evaluate the models, we used several classification metrics rather than relying on only one number. This gives a more complete view of model performance.
Accuracy measures the percentage of total predictions that were correct. It is useful as a general overview but it can sometimes be misleading if the dataset is imbalanced. In our case, the test set was close to balanced between positive and negative reviews so accuracy was useful.

Precision measures how many predicted positive examples were actually positive. In other words, when the model predicts a review is positive, precision tells us how often that prediction is correct.

Recall measures how many actual positive examples the model correctly identified. A model with high recall is good at finding positive reviews and avoids missing many of them.
F1 score is the harmonic mean of precision and recall. It is useful because it balances the two metrics. A model with high precision but low recall, or high recall but low precision may not be reliable overall. F2 score gives us a single value that reflects both.
We also used confusion matrices to show the number of correct and incorrect predictions for each class. A confusion matrix makes it easier to see whether the model makes false positives or false negative errors.
Finally, we used ROC curves. The ROC curve shows the tradeoff between true positive rate and false positive rate across different classification thresholds.

Results

We evaluated all models on the same test set consisting of 600 samples with a balanced distribution of 298 negative and 302 positive reviews. This ensured a fair comparison across all models.

The results showed that all three models achieved strong and similar performance. The CNN and Naive Bayes models both achieved an accuracy of approximately 86.5%, while the Gradient Boosting model achieved approximately 86.3%. The F1 scores were also close across all models, indicating balanced performance between precision and recall.

CNN

Class	Precision	Recall	F1
Negative	0.875	0.849	0.862
Positive	0.855	0.881	0.868
Macro avg	0.865	0.865	0.865

Naive Bayes

Class	Precision	Recall	F1
Negative	0.860	0.869	0.865
Positive	0.870	0.861	0.865
Macro avg	0.865	0.865	0.865

Gradient Boosting

Class	Precision	Recall	F1
Negative	0.880	0.839	0.859
Positive	0.848	0.887	0.867
Macro avg	0.864	0.863	0.863

Conclusion

In this project, we developed and evaluated a sentiment analysis system using multiple machine learning approaches including Naive Bayes, Gradient Boosting, and a Convolutional Neural Network. All models achieved strong performance with accuracy around 86%, demonstrating that sentiment classification can be effectively addressed using both classical and neural methods. Our project shows that sentiment analysis can be effectively solved using multiple modeling approaches. The final system is able to process review tests, predict sentiment, and compare model performance using meaningful evaluation metrics.

Future improvements could include testing our trained models against a different set of reviews to check if its results are truly generalized, larger datasets for a wider breadth of data to train our models more completely, and exploring differing model architectures to examine their performance for this problem.